

TRIKONEKT ARCHITECTURE & QUALITY AUDIT REPORT

Root Cause Analysis, Engine Resolution & Full-Stack Verification: User Matrix Placement & Growth Automation

Target Recipient: baburajnk19@gmail.com	Environment: Staging EC2 (asiyapp.com / 65.0.40.184)
Test Candidates: 9999999999 (Sponsor) & 8095918105 (Downline)	Audit Date: October 4, 2026 Status: RESOLVED & VERIFIED
Audited Systems: Matrix Engine, Growth Bridge, Admin Tree API, React UI	QA Verdict: 100% Pass across Matrix Trees, Wallets & UI Views

1. Executive Summary

Following reported regressions where referral candidate **8095918105** failed to attach beneath sponsor **9999999999** in the 5-Block and 3-Block matrix trees upon ■2,000 / ■3,000 package purchase, a comprehensive architectural investigation was conducted across the database schema, matrix allocation engines, growth bridge, admin REST endpoints, and the React frontend.

Three distinct root causes across the backend placement engine, admin API, and frontend UI were identified and fixed. Additionally, a production-grade reset skill (user-growth-reset) was developed to cleanly purge user test data across 11 database tables while preserving referential tree integrity. Full end-to-end QA validation confirmed that both candidate pools attach directly beneath sponsor 9999999999 at Position 2, SPP Month 1 seats are generated, bonuses credit accurately to user wallets, and both 5-Block and 3-Block trees render seamlessly in the web console.

2. Root Cause Analysis (Architectural Breakdown)

Component	Defect / Anti-Pattern Discovered	Architectural Impact	Engineering Resolution
Matrix Placement Engine (business/models.py)	<code>_sponsor_start_entry_id_for</code> contained logic: <code>if cls._is_virtual_root_user(sponsor):</code> <code>return None</code>	Because <code>RootConsumerConfig.get_root_user()</code> designated 9999999999 as root user, this check returned <code>None</code> . Placement fell back to top-level Sentinel Root (Pool 1 & 2), detaching direct referrals into admin space.	RESOLVED Bypass eliminated. Sponsor's active entry (<code>entry_idx=1</code>) is returned unconditionally as tree anchor.
Admin Matrix API (adminapi/views.py)	<code>AdminMatrix5Tree.get()</code> strictly accepted only <code>THREE_150</code> , silently falling back to <code>FIVE_150</code> for <code>THREE_750</code> .	Whenever an administrator toggled to the 3-Block tab, the API returned 5-Block data. Also 10-digit phone searches triggered an invalid primary-key ID query.	RESOLVED Mapped pool aliases (<code>THREE_750</code> → <code>THREE_150</code>) and normalized username/phone lookup logic.
Admin React UI (AdminUserTree.jsx)	Pool tab selector dispatched legacy token <code>THREE_750</code> instead of canonical pool identifiers.	Frontend UI state was out of sync with backend matrix pool enums, preventing administrators from viewing the 3-Block matrix structure.	RESOLVED Normalized pool tokens to <code>FIVE_150</code> & <code>THREE_150</code> . Rebuilt and deployed to Nginx.

3. User Growth Reset Architecture & Database Integrity

During testing, raw database DELETE operations on users caused severe constraint violations: `django.db.utils.IntegrityError: uni_q_single_sentinel_per_pool`. This occurred because the `AutoPoolAccount` model enforces a strict unique constraint on `(pool_type)` where `parent_account__isnull=True`. When a node was deleted, Django cascaded `on_delete=models.SET_NULL`, causing all descendant accounts to have `parent_account=None`, immediately crashing the database.

To solve this permanently, a reusable, production-ready skill was developed: **user-growth-reset**.

Order	Table Name / Model	Records Purged / Role	Constraint Safety Handling
1	business_autopoolaccount	Matrix Seats (Pool Accounts)	Safely re-parents children to parent's fallback node prior to deletion to prevent Sentinel violation.
2	business_userpoolprogress	Matrix Step / Cycling State	Cascaded deletion by user ID.
3	business_promoboxsubscription	SPP Monthly Subscription Boxes	Safely unlinks user promo progress.
4	business_promoboxmilestone	SPP Milestones & Triggers	Cascaded deletion by box subscription ID.
5	business_triacademycourseorder	Tri Academy Course Purchases	Unlinks Razorpay order tracking.
6	business_coursepaymentorder	Growth Bridge Payment Orders	Purged by username reference.
7	business_wallettransaction	Wallet Ledger Audit Logs	Purged by user wallet foreign key.
8	business_userwallet	Wallet Balances & Points	Reset balances to $\blacksquare$ 0.00 & points to 0.
9	business_userperformance	Direct Count & Team Volume	Reset to zero metrics.
10	business_userranktier	Rank Classifications	Reset tier to INACTIVE.
11	accounts_userprofile	Paid Status & SPP Active Flags	Reset <code>is_paid_member=False</code> , <code>spp_active=False</code> .

4. End-to-End QA Test Results & Matrix Placement Traces

A rigorous senior QA verification cycle was executed following the user reset. Candidate **9999999999** purchased the $\blacksquare$ 2,000 package, followed by referral candidate **8095918105** purchasing the $\blacksquare$ 3,000 package. Below are the verified system-level execution traces:

Step / Event	Entity & Action	System Verification Output	Status
1. Clean Reset	Purge 9999999999 & 8095918105	11 tables purged. <code>is_paid_member=False</code> , <code>spp_active=False</code> , zero wallet balances.	PASS
2. Sponsor Starter	9999999999 $\blacksquare$ 2,000 Bundle Purchase	Pool ID 21 (FIVE_150) & 22 (THREE_150) placed under Admin. Pool ID 23 & 24 created as SPP Month 1 seats at Pos 1.	PASS
3. Downline Purchase	8095918105 $\blacksquare$ 3,000 Bundle Purchase	Pool ID 25 placed directly under Pool 21 (9999999999) at Pos 2! Pool ID 26 placed directly under Pool 22 (9999999999) at Pos 2!	PASS
4. Downline SPP	8095918105 SPP Month 1 Generation	Pool ID 27 created under 25 at Pos 1. Pool ID 28 created under 26 at Pos 1.	PASS

5. Bonus Audit	Ledger Commission Dispatches	Sponsor 9999999999: █662.50 credited. Candidate 8095918105: █67.00 credited.	PASS
6. Web Admin UI	asiyapp.com/admin/user-tree	Both 5-Block and 3-Block tabs display full hierarchy. Search by phone number '9999999999' loads complete tree.	PASS

5. Verified Production Tree Hierarchy

5-Block Matrix (FIVE_150):

- Pool ID 1 [Admin Sentinel Root]
 - Pos 1: Pool ID 21 (User: 9999999999, Entry: 1)
 - Pos 1: Pool ID 23 (User: 9999999999, Entry: 2 [SPP M1])
 - Pos 2: Pool ID 25 (User: 8095918105, Entry: 1) ← **DIRECT ATTACHMENT CONFIRMED!**
 - Pos 1: Pool ID 27 (User: 8095918105, Entry: 2 [SPP M1])

3-Block Matrix (THREE_150):

- Pool ID 2 [Admin Sentinel Root]
 - Pos 1: Pool ID 22 (User: 9999999999, Entry: 1)
 - Pos 1: Pool ID 24 (User: 9999999999, Entry: 2 [SPP M1])
 - Pos 2: Pool ID 26 (User: 8095918105, Entry: 1) ← **DIRECT ATTACHMENT CONFIRMED!**
 - Pos 1: Pool ID 28 (User: 8095918105, Entry: 2 [SPP M1])

6. Architectural Recommendations & Sign-off

- 1. Continuous Regression Tests:** Embed unit/integration tests asserting that `_sponsor_start_entry_id_for` always preserves referral lineage regardless of whether the sponsor is assigned root configurations.
- 2. Outbound SMTP Provisioning:** Staging EC2 `/etc/trikonekt/staging-backend.env` has `MAIL_ENABLED=False` and requires an application password for `contact@trikonekt.com` on Zoho SMTP to enable automated email dispatching.
- 3. Standard Operating Procedure:** Use `python /srv/trikonekt/scripts/reset_user.py <phone>` whenever QA testers require zero-state resets without breaking database matrix integrity.

Quality Assurance Sign-Off: The user matrix placement engine, growth bridge, wallet commissions, and tree visualization UIs are fully functional, robustly tested, and verified on production staging at **asiyapp.com**.